

Day 2 - MLR

$$X = \begin{bmatrix} 1 & 5 & 11 & 125 \\ 1 & 7 & 9 & 155 \\ 1 & 9 & 15 & 175 \\ 1 & 11 & 13 & 220 \\ 1 & 13 & 18 & 235 \end{bmatrix}$$

$$Y = \begin{bmatrix} 48 \\ 72 \\ 101 \\ 142 \\ 176 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 5 & 7 & 9 & 11 & 13 \\ 11 & 9 & 15 & 13 & 18 \\ 125 & 155 & 175 & 220 & 235 \end{bmatrix}$$

$$(X^T X) = \begin{bmatrix} 5 & 45 & 66 & 910 \\ 45 & 445 & 638 & 9515 \\ 66 & 638 & 920 & 13775 \\ 910 & 9515 & 13775 & 225270 \end{bmatrix} = A$$

$$Q = (X^T X)^{-1} X^T Y$$

Gauss elimination method

[A|I]

$$\left[\begin{array}{cccc|ccc} 5 & 45 & 66 & 910 & 1 & 0 & 0 & 0 \\ 45 & 445 & 638 & 9315 & 0 & 1 & 0 & 0 \\ 66 & 638 & 920 & 13775 & 0 & 0 & 1 & 0 \\ 910 & 9315 & 13775 & 225270 & 0 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{cccc|ccc} 1 & 9 & 13.2 & 182 & 0.2 & 0 & 0 & 0 \\ 45 & 445 & 638 & 9315 & 0 & 1 & 0 & 0 \\ 66 & 638 & 920 & 13775 & 0 & 0 & 1 & 0 \\ 910 & 9315 & 13775 & 225270 & 0 & 0 & 0 & 1 \end{array} \right] R_1 \rightarrow R_1/5$$

$$\left[\begin{array}{cccc|ccc} 1 & 9 & 13.2 & 182 & 0.2 & 0 & 0 & 0 \\ 45 & 445 & 638 & 9315 & -9 & 1 & 0 & 0 \\ 66 & 638 & 920 & 13775 & 0 & 0 & 1 & 0 \\ 910 & 9315 & 13775 & 225270 & 0 & 0 & 0 & 1 \end{array} \right] R_2 \rightarrow R_2 - 45R_1$$

eliminating 66 in Row 3

$$R_3 \rightarrow R_3 - 66R_1$$

$$[9, 44, 48.8, 1762] - 13.2, 0, 10]$$

eliminating 910 in Row 4

$$[0, 1325, 1763, 59650 | -182, 0, 0, 1]$$

Matrix become

$$\left[\begin{array}{cccc|cccc} 1 & 9 & 13.2 & 182 & 0 & 12 & 100 & 6 \\ 0 & 40 & 44 & 1325 & -9 & 1 & 0 & 6 \\ 0 & 44 & 48.8 & 1763 & -13.2 & 0 & 10 & \\ 0 & 1325 & 1783 & 59650 & -182 & 0 & 0 & 1 \end{array} \right]$$

$$R_2 \rightarrow R_2/40, R_1 \rightarrow R_1 - 9R_2, R_3 \rightarrow R_3 - 44R_2, R_4 \rightarrow R_4 - 1325R_2$$

$$(X^T X)^{-1} = \begin{bmatrix} -2.49 & -0.929 & 0.005 & -0.0121 \\ -0.929 & 0.0567 & 0.0829 & -0.0037 \\ 0.0052 & 0.0829 & -0.1811 & 0.0035 \\ -0.0121 & -0.0037 & 0.0035 & -0.00048 \end{bmatrix}$$

$$(X^T Y) = \begin{bmatrix} 539 \\ 5503 \\ 7705 \\ 107435 \end{bmatrix}$$

$$Q = (X^T X)^{-1} (X^T Y)$$

$$Q = \begin{bmatrix} -99.9112 \\ 1.2174 \\ 2.4166 \\ 0.9058 \end{bmatrix}$$

$$y = 1.217x_1 + 2.419x_2 + 0.905x_3 - 99.9112$$

$$\text{Wind speed} = 9 \text{ m/s}$$

$$\text{Blade angle} = 11^\circ$$

$$\text{Rotor speed} = 100 \text{ RPM}$$

$$\begin{aligned} \text{Power Output} &= 1.217 \times 9 + 2.419 \times 11 + 0.905 \times 100 - 99.9112 \\ &= \underline{\underline{28.21 \text{ kW}}} \end{aligned}$$